
CSE 144 Final Project — Spring 2026

Ben

Training pipeline implementation,
hyperparameter tuning

Milan

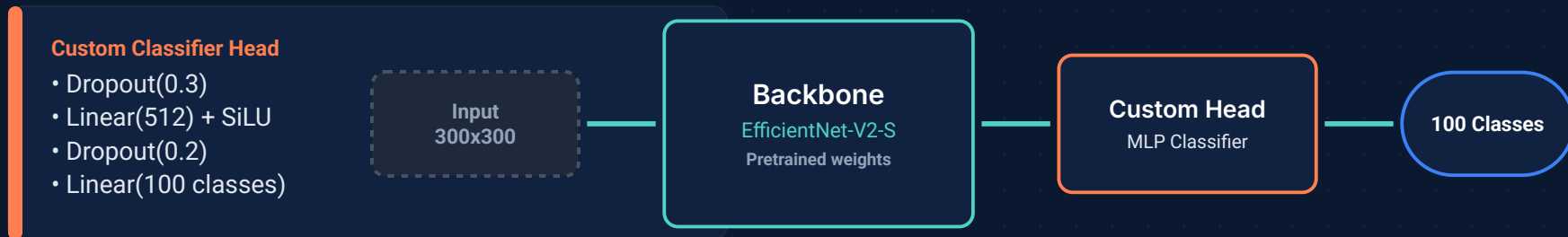
Model architecture design and
classifier head modifications.

Aaron

Data augmentation strategy and
dataset preprocessing.

EfficientNet-V2-S Architecture & Custom Head

- **Backbone:** EfficientNet-V2-S pretrained on ImageNet
- **Input:** 300x300 RGB images



Two-Phase Fine-Tuning Pipeline & Inference TTA

Phase 1

Head Warmup (5 epochs)

- Backbone frozen
- AdamW optimizer (LR=8e-4)
- OneCycleLR scheduler

Phase 2

Full Fine-Tune (up to 45 epochs)

- All layers unfrozen
- Differential LR (BB: 6e-5, Head: 1.8e-4)
- CosineAnnealingLR scheduler

Inference Strategy

Test-Time Augmentation (TTA)

Use of 4 augmented views averaged during inference to enhance final prediction accuracy.

Key Hyperparameters & Augmentation Strategy

Model Configuration

Image size **300x300**

Batch size **12**

Weight decay **7e-4**

Label smoothing **0.03**

Random seed **43**

Pipeline Augmentations



RandomResizedCrop

Scale 0.82-1.0



RandomHorizontalFlip

p=0.5



ColorJitter

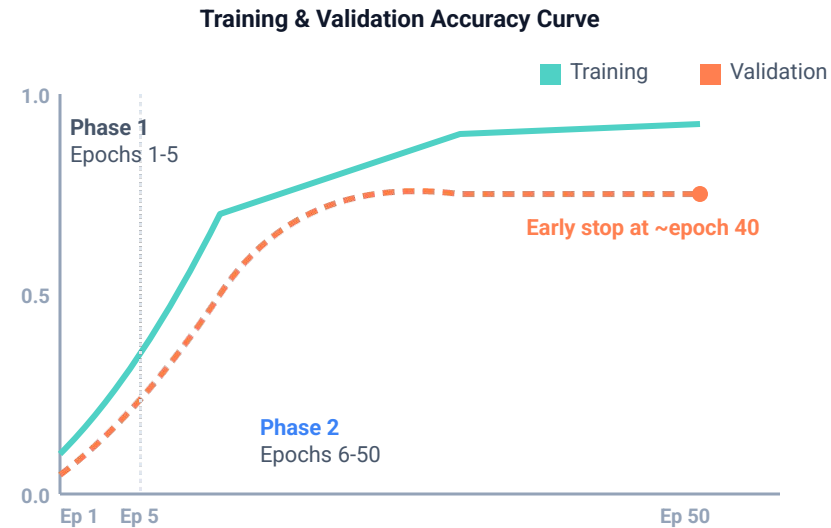
Brightness/Contrast ± 0.12 , Saturation ± 0.08



Results: Kaggle Score & Accuracy

~85%

Public Leaderboard Score



What's Next: Scaling & Future Improvements



Architecture Scaling

Larger backbones (EfficientNet-V2-M, ViT-B/16).



Advanced Augmentation

Enable MixUp, CutMix, or RandAugment.



Resolution

Explore 384x384 input resolution.



Strategy

Ensemble checkpoints for improved stability.



Goal

Push leaderboard score toward 90%+.

Target Metric: 0.90+ Accuracy